PhysBandMamba: A Spectrally Purified State Space Model for Efficient Remote Physiological Measurement

Sacha Chabbert Cherifi[1,*] Haifeng Zhang[2] Von Ralph Dane Marquez Herbuela[2]

[1]ENSEA, Cergy, France
[2]Cognitive Developmental Robotics Lab, International Research Center for Neurointelligence, The University of Tokyo

sacha.cherifi@ensea.fr zhang.haifeng@mail.u-tokyo.ac.jp
herbuela.vonralphdane@mail.u-tokyo.ac.jp

[*]Principal and corresponding author.

Abstract

Remote photoplethysmography (rPPG) recovers the blood volume pulse from facial video. Its natural deployment targets, such as phones, vehicles and embedded monitors, call for models that are compact and that remain reliable when the camera, subjects and recording conditions change. Recent state space models for rPPG pursue accuracy through increasingly elaborate temporal designs: parallel multi-scale paths, bidirectional scans and deep block stacks. We take the opposite direction. **PhysBandMamba** is a deliberately simple, sequential pipeline: a narrow frame stem that fuses each frame with its temporal differences, a fixed, parameter-free spectral purifier that removes the DC component, equalises channel scale and restricts the representation to the physiological band, and two unidirectional Mamba blocks. The model has 0.32M parameters and processes a 160-frame clip in 2.82 ms; in a benchmark of eight architectures measured under one protocol, it is the smallest and the fastest, uses the fewest multiply-accumulates, and is second only to PhysNet in peak memory. It achieves the lowest published error for UBFC→PURE transfer (MAE 0.35, RMSE 0.60 bpm), with an RMSE-to-MAE ratio half that of the strongest baselines, indicating that its advantage comes from avoiding gross subject-level failures. It is competitive on PURE, UBFC-rPPG and VIPL-HR, and we analyse why it trails the best transformer on the mobile MMPD benchmark.

1 Introduction

Photoplethysmography measures blood volume changes optically. Contact sensors are accurate but intrusive; remote photoplethysmography recovers the same pulse waveform from ordinary RGB video of the face [25], enabling unobtrusive monitoring in clinical, automotive and consumer settings. The difficulty is one of signal-to-noise: the pulse is a sub-percent modulation of skin reflectance, buried under illumination change, subject motion, sensor noise and video compression.

Early work approached this with explicit optical models and blind source separation [19, 4, 27, 5, 18]. Deep learning then progressed from 2D CNNs with motion/appearance branches [2, 10, 11] to 3D spatiotemporal CNNs [29], to transformers [31, 30, 34], and most recently to state space models [35]. In-domain accuracy has improved steadily, often through increasingly elaborate temporal modules. Two properties that matter for deployment have improved less consistently: cost, and robustness when the test camera, subjects and protocol differ from training.

This paper asks how simple an rPPG state space model can be while remaining accurate and robust. Our answer rests on an observation about the signal: the human pulse occupies a narrow, known frequency band, roughly 40 to 180 bpm, and energy outside it is by construction not the signal. Rather than relying on a deep temporal model to learn this from the training distribution, we build it into the pipeline with a fixed, parameter-free spectral purifier placed between the frame stem and the temporal core. With the temporal core's input cleaned in this way, the core itself can be minimal: two unidirectional Mamba blocks applied in sequence, without parallel multi-scale paths, bidirectional scans or deep stacks.

The resulting model, PhysBandMamba, is a complete design in which each component is chosen for what it contributes at a small budget: temporal differences in the stem, a factorised widening step, the parameter-free purifier, a shallow sequential core, and a face-separation regulariser that is used only during training and therefore costs nothing at inference. The prior also has a clear limit, which our results reflect: a fixed band cannot separate the pulse from nuisance that lies inside the band, such as the cadence of walking.

Our contributions are:

- PhysBandMamba, a simple sequential architecture of 0.32M parameters that combines a narrow temporal-difference stem with factorised widening, a fixed parameter-free spectral purifier, and two unidirectional Mamba blocks; among eight architectures measured under one protocol, it has the fewest parameters and the shortest inference time.
- Evidence that this simplicity is viable because of the spectral constraint: at fixed depth, removing the purifier mul-

tiplies cross-dataset RMSE by 5.4.

- The lowest published UBFC→PURE error (MAE 0.35, RMSE 0.60 bpm), with an RMSE-to-MAE ratio half that of the strongest baselines, and competitive results on PURE, UBFC-rPPG, VIPL-HR and transfer onto MMPD.

2 Related Work

2.1 Remote Physiological Measurement

Hand-crafted methods recover the pulse from spatially averaged colour traces using optical models of skin reflectance: blind source separation [19], chrominance-based projection [4], plane-orthogonal-to-skin [27], blood volume pulse signatures [5] and local group invariance [18]. They are cheap and interpretable but degrade sharply under motion and illumination change.

Learned methods followed. DeepPhys [2] introduced a motion branch driven by normalised frame differences with an appearance branch supplying spatial attention; TS-CAN [10] added temporal shift modules; EfficientPhys [11] removed the separate appearance stream. PhysNet [29] processes the clip as a spatiotemporal volume with 3D convolutions. Later work introduced temporal-difference convolution [32, 28] and disentangled representations [16]. PhysFormer [31, 30] applies temporal-difference-guided attention over spatiotemporal tokens, and RhythmFormer [34] adds hierarchical temporal attention. Self-attention captures long-range periodicity but scales quadratically with clip length.

2.2 Frequency Priors and Compact Models

Band-pass filtering in the physiological range is standard in rPPG, usually as post-processing of the predicted signal. FreqPhys [20] recently brought this prior inside the network, applying a fixed physiological band-pass together with learnable spectral modulation within a frequency-conditioned diffusion model. Our use of the same kind of prior is different in purpose: rather than strengthening a large generative model, we use it to make a very small discriminative one sufficient.

Two recent works target compactness directly. FactorizePhys [8] replaces attention with a matrix-factorisation module and reaches 52K parameters. ME-rPPG [26] uses a temporal-spatial state space formulation for frame-by-frame inference with 580K parameters. Both reduce cost by redesigning the learned components; we instead keep conventional components and simplify the temporal model by constraining its input.

2.3 State Space Models

State space models offer linear-time sequence modelling. Mamba [7] introduced input-dependent selectivity, making the state transition a function of the input, and has been extended to vision [33, 17, 3], often with bidirectional or multi-directional scans. RhythmMamba [35] applies Mamba to rPPG with multi-temporal constraints over parallel paths and a frequency-domain feed-forward block. Its frame stem, a frame-difference and appearance fusion, is effective enough that we adopt it as our front end; our departure lies in what follows it, where we remove rather than add temporal machinery.

3 Methodology

3.1 Overall Framework

Given a clip $x \in \mathbb{R}^{B\times T\times 3\times H\times W}$, the model predicts a blood volume pulse trace $\hat{y} \in \mathbb{R}^{B\times T}$ through a single sequential path: a frame stem, the parameter-free spectral purifier, attention-weighted spatial pooling, and a two-block Mamba temporal core with a pointwise predictor head (Figure 1).

3.2 Frame Stem

We keep RhythmMamba's fusion stem [35], which combines the raw frame with a four-tap temporal-difference stack, at its original narrow width $C/4 = 24$. RhythmMamba then reaches the working width $C = 96$ with a single $(2, 5, 5)$ spatiotemporal convolution that also halves the temporal length (115K parameters). We factorise this step: a $(2, 5, 5)$ convolution to an intermediate width C_{mid}, which performs the temporal halving $T \to T/2$, followed by a pointwise 1×1 convolution to C. At $C_{\text{mid}} = 72$ this costs 94K parameters, a 19% reduction.

The intermediate width acts as a stability parameter rather than merely a compression knob. At $C_{\text{mid}} = C/2 = 48$ the saving rises to 46%, but the optimum becomes markedly less stable. Among $C_{\text{mid}} \in \{56, 64, 72\}$, swept across the four evaluation directions, 72 gave the most reliable result.

Of the model's 0.32M parameters, the stem accounts for 40% (131K) and the temporal core for 60% (193K). The model's compactness therefore comes mainly from the depth of that core, two Mamba blocks, which the purifier makes viable (Section 4.5).

3.3 Spectral Purifier

Let $Z \in \mathbb{R}^{B\times C\times T'\times H_s\times W_s}$ denote the stem output. The purifier applies, independently per channel and per spatial location, DC removal, RMS normalisation, and a band-pass in the Fourier domain:

$$Z_c = Z - \tfrac{1}{T'}\textstyle\sum_t Z, \tag{1}$$

$$Z_n = Z_c \,/\, \max\!\Big(\sqrt{\tfrac{1}{T'}\textstyle\sum_t Z_c^2 + \epsilon},\ 10^{-4}\Big), \tag{2}$$

$$\hat{Z} = \mathcal{F}^{-1}\big[\mathcal{F}[Z_n] \odot M(f)\big], \tag{3}$$

where $M(f)$ is a fixed mask with full pass-band $[0.65, 3.0]$ Hz (39–180 bpm) and cosine edges over $[0.50, 0.65]$ and $[3.0, 3.30]$ Hz:

$$M(f) = r(f; 0.50, 0.65)\,\big(1 - r(f; 3.0, 3.30)\big), \tag{4}$$

$$r(x; a, b) = \tfrac{1}{2} - \tfrac{1}{2}\cos\Big(\pi\,\mathrm{clip}\big(\tfrac{x-a}{b-a}, 0, 1\big)\Big). \tag{5}$$

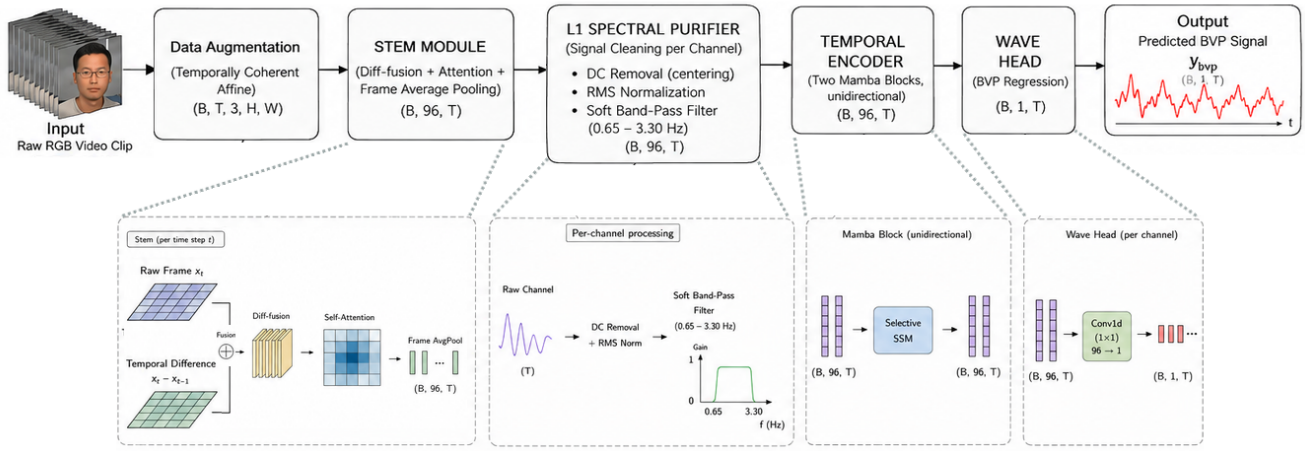


Figure 1: The framework of PhysBandMamba. A frame stem fuses each raw frame with its temporal differences; every channel and spatial location is then cleaned independently by the fixed spectral purifier (DC removal, RMS normalisation, soft band-pass with full pass-band 0.65–3.0 Hz and cosine edges down to 0.50 and up to 3.30 Hz) before attention-weighted pooling; two unidirectional Mamba blocks model the resulting sequence and a pointwise head regresses the pulse.

The purifier operates after the stem's temporal halving, on $T' = 80$ samples at an internal rate of 15 Hz, so its frequency grid has a spacing of 0.1875 Hz (11.25 bpm). At this resolution 13 bins lie in the full pass-band and one bin at each edge is partially attenuated (by factors 0.37 and 0.31); everything else is removed. The soft edges therefore act as a mild taper on the two boundary bins.

The three operations have distinct roles. DC removal discards the static appearance component that dominates the raw features. RMS normalisation puts every channel on a common scale, so that channels dominated by slow illumination change do not overwhelm those carrying the pulse. The band-pass removes what remains outside the physiological range. The module carries **no learnable parameters**: it encodes a physiological fact rather than learning it from one dataset, and it adds no parameters to the model.

Two properties bound what the purifier guarantees. First, the band limit holds per spatial location; the subsequent pooling uses a time-varying attention map, and multiplying by a time-varying weight can reintroduce a small amount of energy outside the band. Second, being a fixed mask, it cannot favour one in-band frequency over another, so in-band confusions, whether a harmonic or a periodic motion, must be resolved by the temporal core.

3.4 Attention, Pooling and Temporal Core

Pooling follows RhythmMamba: a per-frame renormalised map A_{raw} is computed from the sigmoid of the stem output, and $F = \frac{1}{H_s W_s} \sum_{h,w} \hat{Z} \odot A_{\text{raw}}$.

We add one auxiliary regulariser on the stem. A face map is built from the channel-mean *logit* map by spatial z-scoring followed by a capped sharpening temperature, $A_{\text{face}} = \sigma(\tau(m - \mu)/\sigma_s)$, and a binary-entropy penalty on A_{face} (weight 0.005) encourages it to be bimodal. A_{face} does not gate the pooling; its role is to push the stem's features to separate face from background spatially. Since it acts only through the training loss, it can be dropped at inference at no cost. Working in logit space is necessary: channel-averaged post-sigmoid activations stay flat and never separate.

$F \in \mathbb{R}^{B\times C\times T/2}$ is then modelled by two residual blocks applied in sequence, each consisting of a LayerNorm, a *unidirectional* Mamba (d_{state}=16, d_{conv}=4, expand=2), and a gated pointwise MLP. A 1×1 convolution maps to the waveform, upsampled $\times 2$ to restore T. A bidirectional core would add 136K parameters (+42%), and in preliminary experiments on a related architecture it also generalised worse across datasets; we therefore use a single direction.

Deployment. Like other chunk-based rPPG models, PhysBandMamba runs online with a sliding window: the most recent 160 frames (5.3 s at 30 fps) are buffered and the model is re-run at each hop. At 2.82 ms per window, a one-second hop uses well under 1% of the real-time budget on our GPU. The model is not causal at the frame level, since the purifier and the stem's temporal differences use the whole window.

4 Experiment

4.1 Datasets and Metrics

We evaluate on four datasets. **PURE** [22] contains 60 videos of 10 subjects performing six controlled head-motion tasks, recorded at 30 fps. It is a resting protocol, with a median heart rate of 69 bpm and a genuine bradycardic tail reaching the mid-40s. **UBFC-rPPG** [1] contains 42 videos recorded at 30 fps while subjects play a timed arithmetic game, raising heart rate to a median of 103 bpm. **MMPD** [24] contains 660 videos of 33 subjects captured on a mobile phone across four lighting conditions and four activity levels including walking, and is substantially harder than the other two. **VIPL-HR** [15] con-

tains 2,378 videos of 107 subjects recorded with three devices under varying illumination and motion, and is evaluated with a subject-exclusive 5-fold protocol.

For PURE, UBFC and MMPD we follow a subject-held-out split. For VIPL-HR we run the full subject-exclusive 5-fold cross-validation of [15, 31] and report the mean over the five folds, each trained for 10 epochs; the five test folds partition the dataset exactly, and no subject appears in both the training and the test half of any fold. In every protocol we report the last epoch rather than the best, since none of these benchmarks provides a validation split separate from the test set.

Following standard protocol [12], we report Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Pearson correlation (ρ). Heart rate is obtained by concatenating a subject's chunks and taking the spectral peak within [45, 150] bpm. Because RMSE weights large errors quadratically, the ratio RMSE/MAE is a useful diagnostic: it is close to 1 when errors are evenly spread and grows when a few subjects fail grossly.

4.2 Implementation Details

The model is implemented in PyTorch and evaluated using the open-source rPPG toolbox [12] for fair comparison. Videos are cropped to the face, resized to 128×128 and split into 160-frame chunks. Training uses AdamW (learning rate 3×10^{-4}, weight decay 5×10^{-4}) with OneCycle scheduling for 30 epochs at batch size 16, except on VIPL-HR, whose five folds are each trained for 10 epochs because the dataset is an order of magnitude larger. The loss combines a negative-Pearson term and a frequency cross-entropy term with weights 0.2 and 1.0, plus the mask-entropy penalty at weight 0.005. Augmentation comprises per-sample temporal rate resampling and nuisance perturbations (affine, warp). Intra-dataset protocols hold out the last 20% of subjects; cross-dataset evaluation uses the entire target dataset.

4.3 Intra-Dataset Evaluation

Table 1 reports intra-dataset results. On PURE and UBFC-rPPG the model is on par with the strongest published methods at a fraction of their size. On VIPL-HR its mean MAE (4.28) is in the same range as RhythmMamba (4.30) and RhythmFormer (4.51); the spread across our five folds is itself wide, from 3.70 to 4.84 bpm, so we read this as parity rather than as a ranking.

Two caveats apply to the easier benchmarks. First, with the standard Welch settings used throughout this literature the spectral grid spacing is 0.54 bpm, so MAEs around 0.3 bpm are consistent with most test subjects landing on the correct bin and the remainder one bin away; differences of a few tenths of a bpm are at the edge of what the protocol resolves. Second, published baselines use slightly different splits. The more informative comparisons are therefore the harder benchmarks and the cross-dataset transfers, where differences are well above resolution.

MMPD. MMPD is the one benchmark where PhysBandMamba clearly trails (MAE 4.01 against 3.07 for RhythmFormer), and the gap is consistent with the limit of the spectral prior. MMPD includes walking, whose step cadence (typically 1.6–2 Hz) falls inside the pulse band, as well as talking and head rotation with broadband spectra. The purifier passes such nuisance unchanged, so separating it from the pulse falls entirely to a two-block temporal core, whereas RhythmFormer's hierarchical attention has substantially more capacity for this in-band discrimination. The purifier removes what is known to be out of band; it does not, and cannot, resolve in-band confusion.

4.4 Cross-Dataset Evaluation

Table 2 reports cross-dataset transfer across four directions. In each case the model is trained on one dataset and evaluated on the entirety of another, without any adaptation, fine-tuning or target-domain statistics. The two transfers between PURE and UBFC-rPPG probe robustness to a change of recording setup and subject activity, while the two transfers onto MMPD additionally introduce mobile-phone capture, varied lighting and physical activity.

Robustness comes from avoiding gross failures. On UBFC→PURE, PhysBandMamba reaches the lowest published error (MAE 0.35, RMSE 0.60). The RMSE gap is larger than the MAE gap: our RMSE/MAE ratio is 1.7, against 3.5 for RhythmFormer, 3.3 for RhythmMamba and 2.9 for FactorizePhys. The baselines are not uniformly less accurate; rather, a few PURE subjects are mis-estimated by a large margin, and our model avoids those failures.

Why transfer is asymmetric. PURE follows a resting protocol with a median heart rate of 69 bpm and a bradycardic tail, whereas UBFC-rPPG subjects perform a timed arithmetic task at a median of 103 bpm; the two distributions barely overlap. Training on UBFC-rPPG and testing on PURE asks the model to generalise toward rates it has seen less of. A learned temporal model tends to absorb the source range; a fixed band treats 45 bpm and 103 bpm identically, which is consistent with the strong UBFC→PURE result. In the reverse direction, our MAE ties RhythmFormer for second, behind ME-rPPG.

Transfer onto MMPD. Onto MMPD, PhysBandMamba ranks second in MAE from PURE (9.26) and third from UBFC (9.86), ahead of RhythmMamba in both directions and behind RhythmFormer. This mirrors the intra-dataset picture: a change of camera and lighting is largely absorbed, while in-band motion remains the limiting factor.

4.5 Ablation Study

We ablate the purifier together with two components inherited from RhythmMamba whose contribution we wanted to confirm at our much smaller depth: the temporal-difference branch of the frame stem, and the augmentation scheme. Each condition is trained under the same protocol and the same two-block depth on UBFC→PURE (Figure 2, Table 3).

Method	PURE			UBFC-rPPG			VIPL-HR			MMPD		
	MAE	RMSE	ρ	MAE	RMSE	ρ	MAE	RMSE	ρ	MAE	RMSE	ρ
HR-CNN [21]	1.84	2.37	0.98	4.90	5.89	0.64	-	-	-	-	-	-
DeepPhys [2]	0.83	1.54	**0.99**	6.27	10.82	0.65	11.00	13.80	0.11	22.27	28.92	-0.03
PhysNet [29]	2.10	2.60	**0.99**	2.95	3.67	0.97	10.80	14.80	0.20	4.80	11.80	0.60
TS-CAN [10]	2.48	9.01	0.92	1.70	2.72	**0.99**	-	-	-	9.71	17.22	0.44
Gideon et al. [6]	2.30	2.90	**0.99**	1.85	4.28	0.93	9.01	14.02	0.58	-	-	-
Dual-GAN [13]	0.82	1.31	**0.99**	0.44	0.67	**0.99**	4.93	7.68	0.81	-	-	-
PhysFormer [31]	1.10	1.75	**0.99**	0.50	0.71	**0.99**	4.97	7.79	0.78	11.99	18.41	0.18
EfficientPhys [11]	-	-	-	1.14	1.81	**0.99**	-	-	-	13.47	21.32	0.21
NEST [14]	-	-	-	-	-	-	4.76	7.51	**0.84**	-	-	-
Li et al. [9]	0.64	1.16	**0.99**	0.48	0.64	**0.99**	5.19	8.26	0.78	-	-	-
PhysFormer++ [30]	-	-	-	-	-	-	4.88	7.62	0.80	-	-	-
Contrast-Phys+ [23]	0.48	0.98	**0.99**	**0.21**	0.80	**0.99**	-	-	-	-	-	-
RhythmFormer [34]	0.27	0.47	**0.99**	0.50	0.78	**0.99**	4.51	7.98	0.78	**3.07**	**6.81**	**0.86**
RhythmMamba [35]	**0.23**	**0.34**	**0.99**	0.50	0.75	**0.99**	4.30	**7.49**	0.81	3.16	7.27	0.84
PhysBandMamba (ours)	0.27	0.38	**0.99**	0.30	**0.60**	**0.99**	**4.28**	7.67	0.80	4.01	9.57	0.73

Table 1: Intra-dataset evaluation. Baseline figures are as published; train/test splits follow each paper's protocol and may differ slightly from ours. Best in **bold**, second best underlined. Our VIPL-HR figure is the mean over the five subject-exclusive folds.

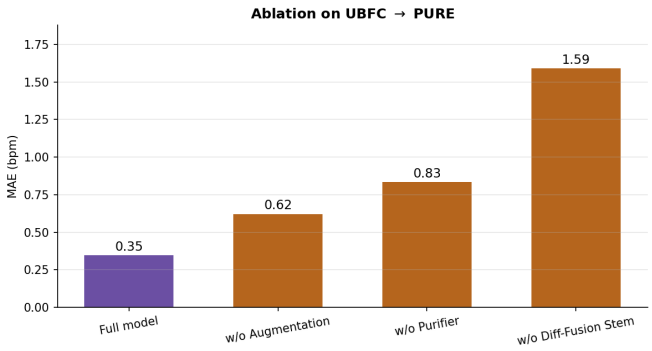


Figure 2: Ablation on UBFC→PURE.

The purifier makes the shallow core viable. Removing the purifier multiplies MAE by 2.4 but RMSE by 5.4, and the RMSE/MAE ratio more than doubles, from 1.7 to 3.9. Without the purifier, most subjects remain accurately estimated but a minority fail badly: the same signature that separates the baselines from our model in Table 2. Because the depth and every other component are held fixed, this also shows that two sequential Mamba blocks are sufficient *with* the purifier but not without it.

The temporal-difference stem remains essential. Removing the temporal-difference branch is the most damaging change (RMSE ×14.5, ratio 5.5): at this depth, the stem's explicit temporal derivatives carry much of the pulse information, and the purifier cannot substitute for them. Removing augmentation raises MAE by 0.27 bpm, about half a spectral bin, with a nearly unchanged RMSE/MAE ratio; its effect is a modest, uniform loss of precision rather than a change in failure mode.

4.6 Computational Cost

Table 4 reports the cost of every model measured under one protocol on the same GPU. Because all figures come from a single benchmark rather than from each paper, they are directly comparable, and the benchmark itself can be checked against the literature: the multiply-accumulate counts it produces reproduce the published cost table of [34] exactly for DeepPhys, PhysNet, TS-CAN, EfficientPhys and PhysFormer, and to within 0.03% for RhythmFormer itself. Parameter counts agree for the four convolutional baselines but not for PhysFormer (7.43M measured against 7.38M published) or RhythmFormer (3.33M against 3.25M), where the released default constructor differs slightly from the configuration each paper evaluated. The gap is largest for RhythmMamba, whose default constructor instantiates a 24-block model at 4.92M parameters against the 1.07M its paper reports. In every case we report what we measured from the released code; against RhythmMamba's published figure, PhysBandMamba is still 3.3× smaller.

Within this benchmark PhysBandMamba is the cheapest model on four of the five axes: it has 2.4× fewer parameters than the next smallest, the fewest multiply-accumulates, 2.8× the throughput of the fastest baseline, and correspondingly the shortest inference time. Peak memory is the one exception, where PhysNet is 51 MB lower; the two are within 12% of each other and both sit far below the remaining methods.

The multiply-accumulate count deserves a specific reading. Both Mamba models sit near 12–13 GMACs because computation is dominated by the shared frame stem operating at 128×128: in our model the stem accounts for more than 99% of all multiply-accumulates, and the temporal core for less than 0.2%. The 7× latency gap between PhysBandMamba and RhythmMamba therefore does not come from fewer operations. It comes from temporal depth: two sequential blocks against twenty-four, each a sequential scan whose cost in time is poorly captured by its operation count. This is precisely where a simple sequential core pays off.

Method	UBFC → PURE			PURE → UBFC			UBFC → MMPD			PURE → MMPD		
	MAE	RMSE	ρ	MAE	RMSE	ρ	MAE	RMSE	ρ	MAE	RMSE	ρ
GREEN [25]	10.09	23.85	0.34	19.73	31.00	0.37	21.68	27.69	-0.01	21.68	27.69	-0.01
ICA [19]	4.77	16.07	0.72	16.00	25.65	0.44	18.60	24.30	0.01	18.60	24.30	0.01
CHROM [4]	5.77	14.93	0.81	4.06	8.83	0.89	13.66	18.76	0.08	13.66	18.76	0.08
LGI [18]	4.61	15.38	0.77	15.80	28.55	0.36	17.08	23.32	0.04	17.08	23.32	0.04
PBV [5]	3.92	12.99	0.84	15.90	26.40	0.48	17.95	23.58	0.09	17.95	23.58	0.09
POS [27]	3.67	11.82	0.88	4.08	7.72	0.92	12.36	17.71	0.18	12.36	17.71	0.18
DeepPhys [2]	5.54	18.51	0.66	1.21	2.90	0.99	17.50	25.00	0.06	16.92	24.61	0.05
PhysNet [29]	8.06	19.71	0.61	0.98	2.48	0.99	9.47	16.01	0.31	13.94	21.61	0.20
TS-CAN [10]	3.69	13.80	0.82	1.30	2.87	0.99	14.01	21.04	0.24	13.94	21.61	0.20
PhysFormer [31]	12.92	24.36	0.47	1.44	3.77	0.98	12.10	17.79	0.17	14.57	20.71	0.15
EfficientPhys [11]	5.47	17.04	0.71	2.07	6.32	0.94	13.78	22.25	0.09	14.03	21.62	0.17
RhythmFormer [34]	0.97	3.36	0.99	0.89	1.83	0.99	**9.08**	**15.07**	**0.53**	**8.98**	**14.85**	0.51
RhythmMamba [35]	1.98	6.51	0.96	0.95	1.83	0.99	10.63	17.14	0.34	10.44	16.70	0.36
FactorizePhys‡ [8]	0.48	1.39	**1.00**	1.04	2.44	0.99	-	-	-	-	-	-
ME-rPPG‡ [26]	-	-	-	**0.53**	**0.85**	**1.00**	-	-	-	10.50	15.70	**0.54**
PhysBandMamba (ours)	**0.35**	**0.60**	**1.00**	0.89	1.77	**1.00**	9.86	16.72	0.35	9.26	16.01	0.39

Table 2: Cross-dataset evaluation, organised by transfer direction. Baseline figures are as published. Best in **bold**, second best underlined. Hand-crafted methods are training-free, so their MMPD figures are identical in both columns. Where two papers report different figures for the same baseline, we take the one less favourable to us: PhysNet onto MMPD is 9.47/16.01 in [35] but 10.24/16.54 in [34], and we use the former. ‡Recent compact models; figures from their own papers, whose evaluation pipelines may differ in detail (FactorizePhys reports ρ of 0.998 and 0.989, averaged over runs).

Configuration	MAE	RMSE	RMSE/MAE
Full model	**0.348**	**0.597**	**1.72**
w/o Augmentation	0.622	1.210	1.95
w/o Purifier	0.833	3.218	3.86
w/o Diff-Fusion Stem	1.593	8.677	5.45

Table 3: Ablation on UBFC→PURE (bpm), all at the same two-block depth. The ratio RMSE/MAE grows when errors concentrate on a few grossly mis-estimated subjects.

Method	**Params** (M) ↓	**MACs** (G) ↓	**Thr.** (Kfps) ↑	**Time** (ms) ↓	**Mem.** (MB) ↓
DeepPhys [2]	1.97	119.1	7.9	20.23	2572
PhysNet [29]	0.77	70.1	20.3	7.89	**433**
TS-CAN [10]	1.97	119.1	7.2	22.20	2678
PhysFormer [31]	7.43	50.6	15.7	10.22	623
EfficientPhys [11]	1.91	59.8	12.1	13.23	1700
RhythmFormer [34]	3.33	38.5	16.3	9.84	611
RhythmMamba† [35]	4.92	12.9	8.1	19.75	490
Ours	**0.32**	**12.4**	**56.7**	**2.82**	484

Table 4: Computational cost, all figures measured under one protocol on the same GPU. Best in **bold**, second best underlined.

All figures are for one 160-frame clip at 128 × 128 in fp32; MACs are multiply-accumulate operations per clip; time is the median of 50 timed runs after 10 warm-up iterations; memory is peak allocation for that clip. †Released default constructor (24 blocks); the paper reports 1.07M parameters.

We do not claim the smallest rPPG model in absolute terms. FactorizePhys [8] reports 52K parameters, and ME-rPPG [26] reports frame-wise CPU inference within a few megabytes of memory; neither was part of our benchmark, and we quote their own figures. Our claim concerns the trade-off: at 0.32M parameters, PhysBandMamba gives the lowest UBFC→PURE error of any published model, including both.

5 Limitations and Future Work

Limitations. The ablation is reported on a single transfer direction, UBFC→PURE. Inference is chunk-based: the model runs online with a sliding window but is not causal at the frame level, and timings were measured on GPU with the CUDA selective-scan kernel rather than on a mobile device. The fixed band assumes adult heart rates between roughly 40 and 180 bpm, and would need to be widened for neonates or intense exercise. Finally, the model trails transformer baselines on MMPD, where periodic motion such as walking falls inside the pulse band and therefore passes through the purifier unchanged.

Future work. Four directions follow directly from these limits. *(i) Frame-level streaming.* Replacing the Fourier-domain mask with a fixed recursive (IIR) band-pass, the chunk statistics with running estimates, and the symmetric temporal differences with causal ones would make every stage causal, letting the Mamba core run in its recurrent mode one frame at a time. *(ii) In-band motion.* A motion reference, such as the spectrum of facial landmark trajectories, could drive an adaptive notch that removes gait and head motion inside the band, addressing the MMPD gap without enlarging the temporal core. *(iii) Beyond heart rate.* The same principle extends to other vital signs with known frequency ranges: a second, lower band (0.1–0.5 Hz) would target respiration with no additional parameters. *(iv) On-device deployment.* Since the stem dominates computation, a lighter or lower-resolution stem, a kernel-free implementation of the selective scan and int8 quantisation are the natural

levers for phones and embedded boards. More broadly, the purifier is independent of the backbone, and placing it in front of other compact architectures would establish how general the observation that a simple temporal core suffices really is.

6 Conclusion

PhysBandMamba shows that an rPPG state space model does not need elaborate temporal machinery to be accurate and robust. A narrow temporal-difference stem, a fixed parameter-free spectral purifier and two unidirectional Mamba blocks in sequence give a model of 0.32M parameters that runs a 160-frame clip in 2.82 ms, is the smallest and fastest in our eight-model benchmark, and achieves the lowest published UBFC→PURE error. Its advantage is concentrated where the spectral constraint applies: it avoids the gross, out-of-band failures that dominate the cross-dataset RMSE of larger models, and the ablation shows that the shallow core is sufficient only with the purifier in place. Where nuisance lies inside the band, as with walking on MMPD, capacity still matters. Deciding which part of the problem can be fixed by design, and learning only the rest, is a cheap and general recipe for compact physiological sensing.

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